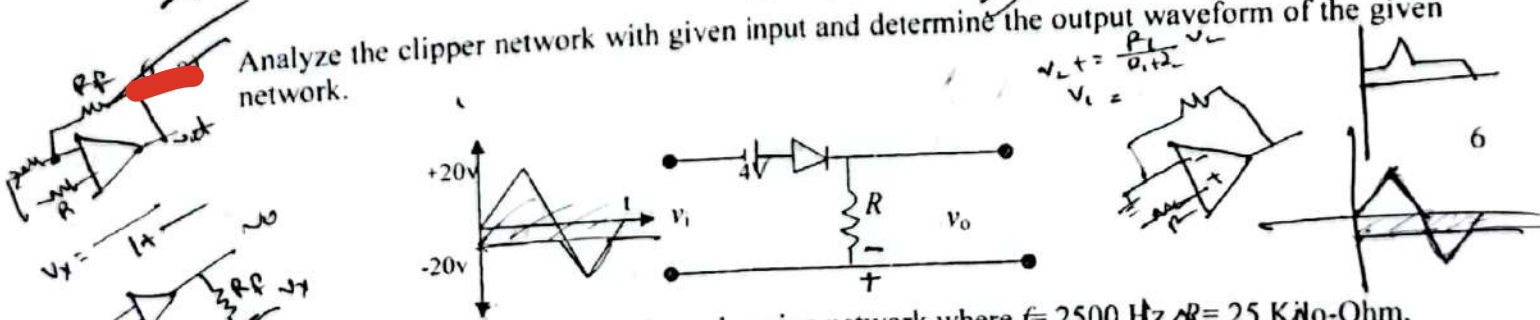


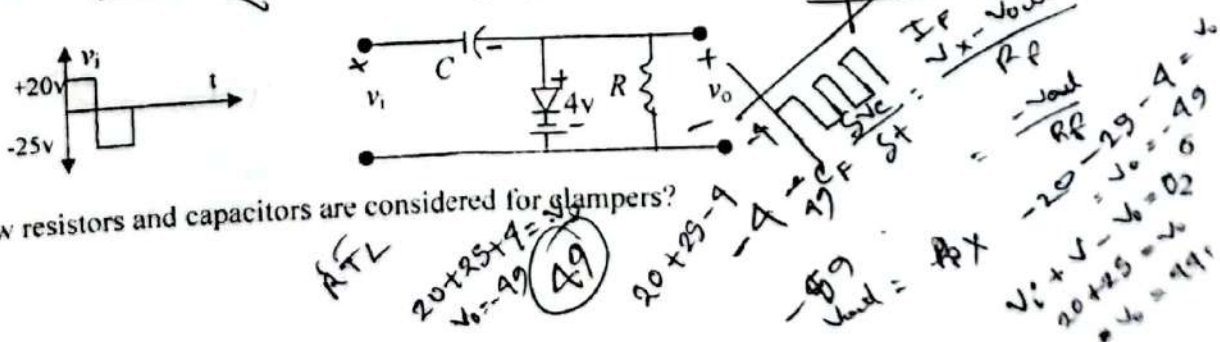
[Figures in the right margin indicate full marks. Split answering of any question is not recommended]
 Answer any 5 of the following questions

1. (a). Explain how transistor is used as inverter? Draw the circuit of a CMOS switch. 3+
- (b). Discuss on speed of operation, noise immunity and figure of merit in digital logic families. 2
- (c). Draw the circuit diagram of 3 inputs RTL NOR gate and explain how it works. 3
2. (a). For the DTL NAND gate where diode and transistors parameters are given as 6
 Diode: Voltage across the diode=0.7V; Cut-in voltage=0.6V
 Transistor: Cut-in voltage=0.5V; $V_{BE,sat}=0.8V$; $V_{CE,sat}=0.2V$; $h_{FE}=20$; $R_C=2\text{ K}\Omega$; $R_B=4\text{ K}\Omega$
 Resistor connected with $V_{CC}=4\text{ K}\Omega$. Calculate i) fan-out ii) noise-margins iii) average power- P . 4
- (b). How 3-input TTL NAND gate is operated in digital logic families? 4
- (c). Discuss about the saturated bipolar logic families. 4
3. (a). Show the pin configuration of a typical op-amp with its function. 6
- (b). Draw the circuit of an Op-Amp as voltage follower and find an expression for its voltage gain. 4
- (c). Draw the circuit for a non-inverting amplifier and explain its operations. 3
4. (a). Show the differences between open loop gain and close loop gain. 6
- (b). Draw the circuit diagram of integrator and differentiator and explain how it works. 5
- (c). Draw the circuit diagram of subtractor and explain how it works. 5
5. (a). Find the analog output voltage of a 3-bit D/A converter for all possible inputs. Assume $K=1$ 5
- (b). Design a weighted-resistor D/A converter and explain how it works. 5
- (c). How offset voltage is used in D/A converter? 2
- (d). What is quantization error? 2

Analyze the clipper network with given input and determine the output waveform of the given network.



Analyze output voltage for given clamping network where $f=2500\text{ Hz}$, $R=25\text{ K}\Omega$ -Ohm, $C=10\text{ nF}$?

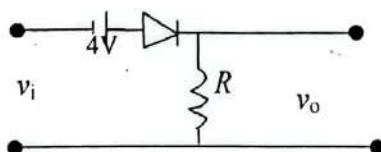
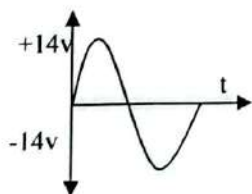


How resistors and capacitors are considered for clampers?

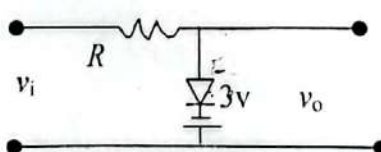
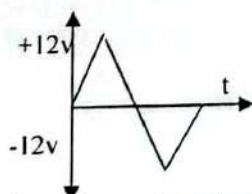
[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions

1. (a) Discuss the switching characteristics of a semiconductor diode. 4
 (b) Draw the circuit diagram of 3-input RTL NOR Gate. Write down the working principle of the 3-input RTL NOR Gate. 6
 (c) Calculate the noise margin of the RTL NOR gate if the parameters are considered as $V_{cc}=5.0$; $R_L=360\Omega$; $R_B=450\Omega$; Cut in voltage of transistor $=0.5\text{ V}$; $N=5$; $h_{FE}=10$. 4
2. (a) Draw the circuit of a 3-input TTL NAND gate and explain how it works. 6
 (b) Show the working principle and applications of CMOS inverter. 4
 (c) Discuss the speed of operation and fan out of logic gates. 4
3. (a) Show the circuit symbol and terminals of operational amplifier 741C. 4
 (b) Show the block diagram for using op-amp in sensor based data acquisition. 4
 (c) Draw the circuit for an inverting amplifier. Deduce the equation of output voltage, current and close loop gain for inverting amplifier. 6
4. (a) Explain how an operational amplifier can be used to design voltage follower? 4
 (b) Show the circuit diagram of integrator and differentiator and explain how it works. 6
 (c) How an operational amplifier can be used to build inverting adder? 4
5. a) Design a 4 bit unipolar D/A converter with $V(1)=-1$, $V(0)=0\text{V}$ and $R_F=8R$. Find the analog output voltage of the D/A converter for each of the digital inputs from 0000 to 1111. 5
 b) Design a 3 bit parallel-comparator A/D converter and explain how it works. 5
 c) What is quantization error? How it is reduced in A/D converter? 4
6. a) Analyze the clipper network with given input and determine the output waveform of the given network. 6



- b) Determine the output waveform step by step of the given network.

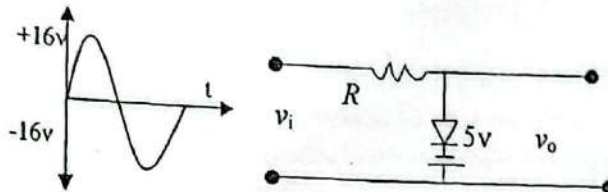


- c) Design the clamping network and show the analysis of the network.

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

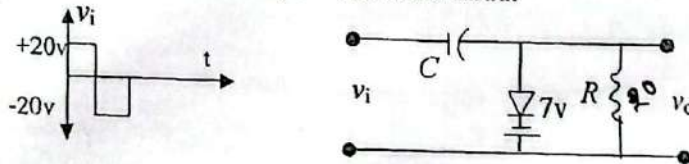
Answer any 5 of the following questions

1. (a). Discuss the bipolar logic families for electronics devices. 4
- (b). Draw the circuit diagram of 3-input DTL NAND Gate. Calculate fan out, noise margin and average power if the parameters are considered as voltage across a conducting diode = 0.7 V; Cut in voltage of diode = 0.6 V; Cut in voltage of transistor = 0.5 V; $V_{BE,sat} = 0.8$; $V_{CE,sat} = 0.2$; $h_{FE} = 20$. 7
- (c). Show the limitations of DTL and TTL circuits. 3
2. (a). Draw the circuit of a TTL gate with totem-pole output driver. Write down the working principle of the circuit. 6
- (b). Determine the output waveform step by step of the given network. 6



- (c). How figure of merit is measured for a digital IC? 2

3. (a). Analyze output voltage and time constant is analyzed in given clamping network where $f = 2000$ Hz, $R = 20$ Kilo-Ohm, $C = 100$ nano-farad. 5



4. (a). What are differences between open loop gain and close loop gain of operational amplifier? 3
- (b). Show how can an operational amplifier be used as an adder and subtractor? 6
5. (a). Explain how an operational amplifier can be used as an integrator? 5
- (b). Show how a low pass filter is designed using the operational amplifier? 5
- (c). How voltage gain is obtained for inverting amplifier? 4

6. (a). Draw the circuit diagram of a multichannel amplifier. Show the output voltage analysis and circuit diagram of differentiator. 6

- (b). Explain how an op-amp can be used as voltage follower? 4
- (c). Give the examples where D/A and A/D converters are used. 4

7. (a). Deduce the output voltage for the weighted-resistor D/A converter. Why offset voltage is used for D/A converter? 3

- (b). Find the analog output voltage of a 5-bit D/A converter where $K=1$. Design the D/A converter for this. 2

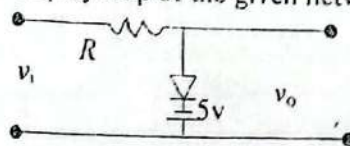
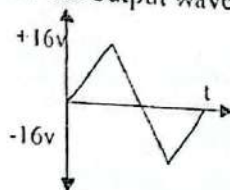
- (c). How 4 bit digital output can be obtained from an analog voltage in the range of 0 to V ? 6

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

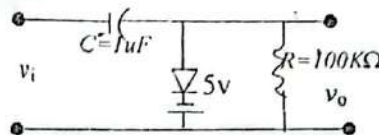
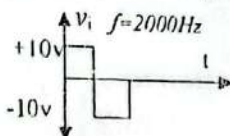
Answer any 5 of the following questions

1. (a). Write down the short note on i) Figure of Merit ii) Fan out iii) Noise immunity. 6
 (b). Draw the circuit diagram of 3-input DTL NAND Gate. Explain the operation of DTL NAND gate. 5
 (c). Show the limitations of RTL and DTL circuits. 3

2. (a). Why clipper is used in electronic circuits? 2
 (b). Determine the output waveform step by step of the given network



- (c). Analyze output voltage and time constant is analyzed in given clamping network? 6



3. (a). Write down the characteristics of ideal op-amp. 3
 (b). What is open-loop voltage gain and closed-loop gain? Show the pin configuration of a typical op-amp. 5
 (c). How output voltage is calculated for the circuit of (i) non-inverting amplifier ii) summing amplifier? 6

4. (a). How cascaded op-amp circuit is designed? How output voltage is obtained in cascaded op-amp circuit? 7
 (b). Design an op-amp circuit with inputs v_1 and v_2 such that $v_o = -7v_1 + 3v_2$. 4
 (c). How voltage gain is obtained for inverting amplifier? 3

5. (a). Show the output voltage analysis and circuit diagram of differentiator. 5
 (b). Explain why capacitor and inductor are very useful in electric circuits. 4
 (c). Design an analog computer circuit to solve the differential equation

$$\frac{d^2 v_o}{dt^2} + 3 \frac{dv_o}{dt} + 2v_o = 4 \cos 10t, t > 0 \text{ subject to } v_o(0) = 2, v_o'(0) = 0$$
 5

6. (a). Analyze the output voltage and K factor of the weighted-resistor D/A converter. 4
 (b). Consider a 4bit D/A converter with $V(1) = -1V$, $V(0) = 0V$ and $R_F = 16R$. Obtain the analog output voltage for each digital input where $V_0 = 0V$ for the digital input of 1000. Draw the circuit of D/A converter and adjust the offset voltage for the circuit. 6
 (c). What is quantization error? How quantization error can be reduced in analog to digital converter? 4

1302059

E3

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam. of B. Sc. Engg. (CSE), July -Dec. 2016

Course Code: EEE 321 Course Title: Digital Electronics and Pulse Technique

Credit Hour: 3.0 Full Marks: 70 Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

- ① (a). Draw a 3-input DTL NAND gate. Show the performance of DTL NAND gate based on your drawn circuit. [Considering standard values of diode and transistor parameters; $h_{FE}=40$] 6
- (b). Discuss about the saturated bipolar logic families. 4
- (c). Find out the logic 0 and logic 1 noise margins for the given circuit diagram of Fig. 1(c) where $h_{FE}=10$ and $N=5$. 4

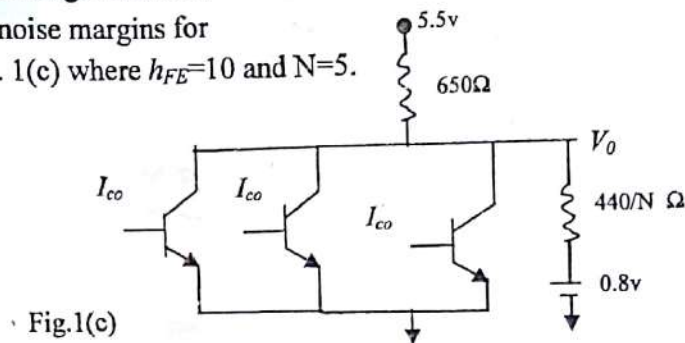


Fig.1(c)

- ② (a). Why TTL is the most popular general purpose logic family? 3
- (b). If the digital input to the D/A converter is $\bar{b}_3b_2b_1b_0$, find the analog output voltage such that $V=0$ for a digital input of 0000. Design a D/A converter for the mentioned criteria. 3+4
- (c). How quantization error can be reduced in A/D converter? 4

- ③ (a). Determine V_O for the given network of Fig. 3(a). 6

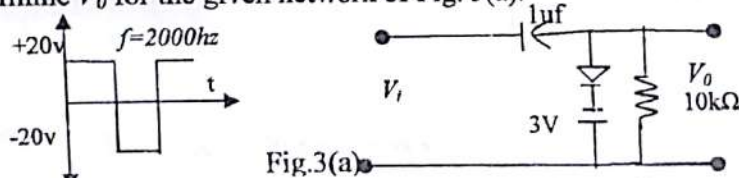


Fig.3(a)

- (b). Determine V_O for the given network in Fig. 3(b) using a silicon diode. 5

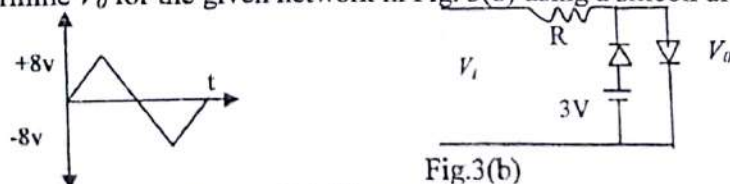
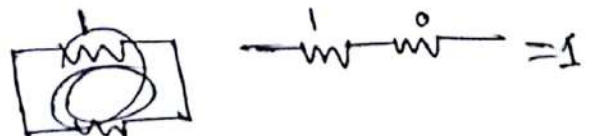


Fig.3(b)

- (c). Distinguish series and parallel clippers. 3

- ④ (a). What is an operational amplifier? "The inverting amplifier provides a constant voltage gain with 180° phase shift"- justify the statement with relevant circuit diagram. 5

- (b). Define slew rate and active device. Solve the differential equation using appropriate circuit diagram: $D^3y - 4D^2y + 6Dy - 8y = f(t)$ volts where $D = d/dt$ 5



Calculate the output voltage (V_{out}) for the circuit of fig.4(c)

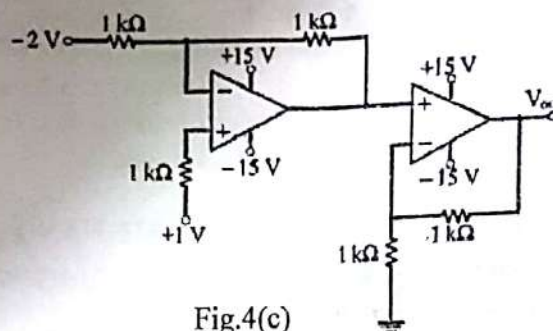


Fig.4(c)

5. (a). Write down the advantage of active filter. Design a 70 kHz with 50% duty cycle square wave generator using 555.

Sketch the equivalent circuit of an ideal Op-Amps. Suppose for a non-inverting amplifier, the equivalent circuit of an Op-Amps is shown in figure 5(b). Find: i) V_s/V_{out} for $R_{in}=100k\Omega$.
ii) V_s/V_{out} for an ideal op-amp.

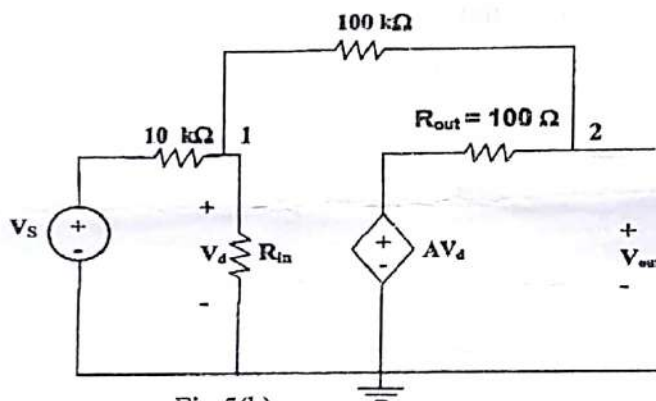


Fig.5(b)

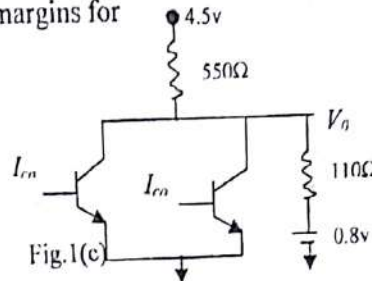
6. (a). What do you mean by cut off frequency? Show that in case of 1st order high pass filter, the transfer function is equal to the 70.7% of it's voltage gain. 5
(b). "Monostable multivibrator is known as one-shot circuit" – Justify the statement with relevant circuit diagram. 4
(c). Explain the functional block diagram of 555 timer. 5

WAN

A	B	C	OU
1	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

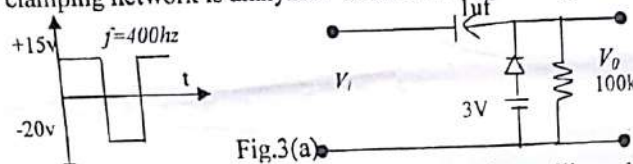
3

1. (a). Explain the operation of a 3-input RTL OR gate with its circuit diagram.
 (b). Write down the short note on i. Figure of merit ii. Noise Margins.
 (c). Find out the logic 0 and logic 1 noise margins for the given circuit diagram of Fig. 1(c).

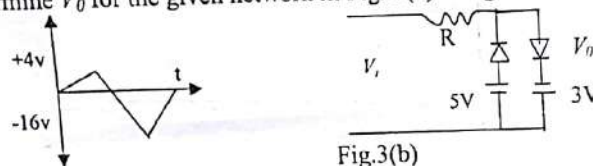


2. (a). Derive the equation of V_0 for weighted resistor D/A converter.
 (b). What are the problems of weighted resistor D/A converter?
 (c). Consider a 3-bit digital input in 1's complement format. Design a D/A converter for this.

3. (a) How clamping network is analyzed? Determine V_0 for the given network of Fig. 3(a).



- (b). Determine V_0 for the given network in Fig. 3(b) using a silicon diode with $V_T = 0.7V$;



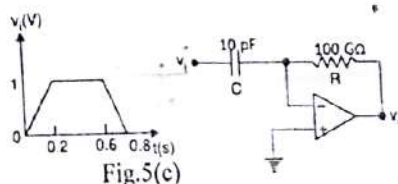
4. (a). What is an operational amplifier? What are the fundamental differences between ideal & practical Op-Amp (Model 741c)?

- (b). Solve the differential equation using appropriate circuit diagram:
 $D^3y + 4D^2y - 6Dy - 8y = -5$ volts where $D = \frac{d}{dt}$

- (c). Write down the short notes on: i. Op-Amp is a subtractor ii. Frequency scaling iii. Schmitt trigger.

5. (a). Design a 50 kHz, 70% duty cycle square wave generator using 555 timer.
 (b). Show that in case of 1st order low pass filter, the transfer function is equal to the 70.7% of it's voltage gain.

- (c). For the differentiator circuit in the Fig. 5(c), determine the output voltage and sketch the output waveform.



6. (a). Evaluate the time period for a free running multivibrator & show that the frequency is independent of output voltage of the square wave.

- (b). "Monostable multivibrator is known as one-shot circuit" – Justify the statement with relevant circuit diagram.

- (c). Explain the functional block diagram of 555 timer.

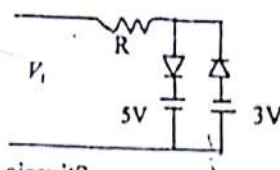
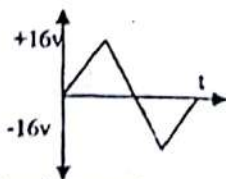
14

E3

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

1. (a). Explain the operation of a 3-input TTL NAND gate with its circuit diagram. 6
(b). Explain the operation of 2-input CMOS OR gate 4
(c). In the RTL 2-input NOR gate, $V_{cc} = 5V$, all resistances are 640Ω . Calculate the average power supply when it is driving 4 gates. Assume $V_{BE, Sat} = 0.8V$, $V_{CE, Sat} = 0.2V$, $h_{FE} = 20$. 4
2. (a). Discuss the operations of a monostable multivibrator with its circuit diagram where a trigger input is given at a NAND gate. 7
(b). Draw a circuit diagram for a comparator where reference voltage $V_{ref} = 5V$ and input voltage $V_{in} = 10 \sin 4\pi t$. Show the input and output waveforms for the given circuit. 7
3. (a). Draw the functional block diagram of 555 timer. Explain its operations. 8
(b). Discuss the function of a monostable multivibrator using 555 timer 6
4. (a). Design a D/A converter with $V(1) = -2V$, $V(0) = 0V$ and $R_F = 4R$ for 5 bit digital input from 00000 to 11111 7
(b). Adjust the offset voltage for the D/A converter such that $V_0 = 0$ for digital input of 01000. 7
5. (a). Design a 3-bit parallel comparator A/D converter. Explain the operation of this A/D converter. 8
(b). Explain how the quantization error can be reduced for this A/D converter. 6
6. (a). Determine V_0 for the given network using a silicon diode with $V_T = 0.7V$; 6



- (b). Why clamper is used in the electronic circuit?
Determine V_0 for the given network

